

Roberto Olvera-Hernandez

Ph.D. student — Biomedical Sciences & Computational Genomics

✉ roberto.olvhez@gmail.com ☎ (+52) 33-1540-9988
in [rob-olvhez](#) 🔗 [robolvhez](#)

🎓 Academic career



Ph.D. in the Doctoral Program of Biomedical Sciences (PDCB)

Center for Genomic Sciences (CCG) — UNAM ([Sohail Lab](#))

Feb. 2025 - ongoing

Cuernavaca, Morelos, MX.

- **Research project:** “Genome-wide Association Studies free of population confounds for complex traits and diseases for over 100 thousand Mexicans”
- **Supervisor:** Dr. Mashaal Sohail
- **Collaborators:** Dr. Jason Torres, Dr. Alexander T. Young



B.Sc. in Biotechnology Engineering

Instituto Tecnológico y de Estudios Superiores de Occidente (ITESO)

2018 - 2023

San Pedro Tlaquepaque, Jalisco, MX.

Research placements



MCPS — Research Placement

Nuffield Department of Public Health (NDPH) — University of Oxford

Aug. 2025 - Jan. 2026

Oxford, Oxfordshire, U.K.

Developed a robust within-family GWAS pipeline for MCPS, enabling unbiased heritability and genetic correlation estimates for 15 complex traits—directly addressing confounding from population structure and shared environment in a highly admixed cohort.

- **Supervisors:** Dr. Jason Torres, Dr. Diego Aguilar-Ramirez



Bachelor thesis

Institute of Biotechnology (IBt) — UNAM

Aug. 2023 - Dec. 2023

Cuernavaca, Morelos, MX.

- **Thesis title:** “Genome-Wide Association Study (GWAS) for the identification of genes related to hydrotropic response in native maize (*Zea mays* L.) from Mexico.”
- **Supervisor:** Dr. Gladys Iliana Cassab López

👜 Professional career



Income Analyst

TOKA Internacional

2024 - 2025

Zapopan, Jalisco, MX.

- Income monitoring and diagnosis within the *fintech* market · Development of SQL/R scripts for massive data analysis · Generation and updating Power BI productive dashboards · Mapping and standardization of management processes such as billing.



Animal Health Diagnostics Trainee

Boehringer Ingelheim — Vetmedica

2021 - 2022

Zapopan, Jalisco, MX.

- NGS protocols for animal viral surveillance · Implementation of Power BI for the marketing team · Document automation using AI · Serological and molecular biology diagnostics · Lab supply management.

Meetings and Conferences



XX Congreso Nacional de Biotecnología y Bioingeniería
Mexican Society of Biotechnology and Bioengineering (SMBB)
May. 2023

- **Short communication:** <https://smbb.mx/revista-biotecnologia-2023-vol-27-no-4-2/>



BerryVax (ITESO) - Online Jamboree
iGEM Design League
Nov. 2021

- **Project presentation:** “Use of *Bacillus subtilis* as biological machine for easy detection and elimination of plagues in berries”
- **Awards:** Silver Medal · Best Presentation

Courses and certifications

Course - Introduction to Epidemiology

Nuffield Department of Public Health (NDPH) - University of Oxford
Jun. 2026 -

Ph.D. Selected Topic — “Clinical, Genetic, and Molecular Aspects of Obesity, Type 2 Diabetes, and Metabolic Syndrome. From the Clinic to Omics Sciences”

Unidad de Investigación Médica en Bioquímica. Unidad Médica de Alta Especialidad “Dr. Bernardo Sepúlveda”, Centro Médico Nacional Siglo XXI, IMSS, CDMX.
Feb. 2025 - May. 2025

Course - “Bioinformatic Tools for High-Throughput Sequencing”

USMBB - IBt UNAM
Jul. 2023

Teacher Assistant in the *Programming with R* workshop.

Course - “Synthetic Biology and Genetic Circuits 101”

Scintia
Jul. 2021

Workshop - “Malice Analysis for your Project”

Engineering Biology Research Consortium (EBRC)
Feb. 2021

Course - “Agile Process, Project, and Program Controls”

Engineering Biology Research Consortium (EBRC)
Feb. 2021

Data Science: R Basics

Engineering Biology Research Consortium (EBRC)
Aug. 2020

Languages

Spanish (native)

Writing
Reading
Conversation



English (TOEFL iBT - C1)

Writing
Reading
Conversation

